

Aishik Mandal

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🎓 EDUCATION

- 09/23 – 12/26 **ELLIS PhD Candidate in Computer Science**, Technical University of Darmstadt
Research: Privacy-Preserving High-Fidelity Synthetic Therapy Session Generation
Advisors: Prof. Iryna Gurevych, Prof. Aurélien Bellet, Prof. Tanmoy Chakraborty
- 2018 – 2023 **Integrated Dual Degree in AI/Electronics**, IIT Kharagpur CGPA: 9.61/10
B.Tech in Electronics and Electrical Communication Engineering
M.Tech in Artificial Intelligence, Machine Learning and Applications
Minor in Computer Science and Engineering
Micro Specialisation in Embedded Control, Software, Modelling and Design

💼 WORK EXPERIENCE

- 10/25 – 03/26 **Visiting Researcher** at **Premedical Lab** under Prof. Aurélien Bellet, INRIA Montpellier, France
Privacy-Utility Trade-offs in Synthetic Therapy Session Generation
- Analyzed the privacy-utility trade-off between using different levels of private data (demographics, background situation, symptoms) as inputs to synthetic therapy session generation pipelines.
 - Found private information like demographics and background situation raise privacy risk by **+52.8% attribute leakage**, **+37.7% linkage attack success**, **+27.2% membership inference**, but improve diversity.
 - Designed noise-based defenses that **reduced attribute leakage by 35% and linkage/membership-inference risk by 20% each**, preserving generation utility.
- 05/22 – 07/22 **Research Intern** at **Articulab, Cognitive Machine Learning Group** under Prof. Justine Cassell, INRIA Paris, France
Multimodal Turn Taking in Conversational Agents
- Developed a multimodal turn-taking predictor combining LSTMs, attention, and a pretrained transformer over audio, visual, and text signals and achieved a **+0.09 macro-F1** improvement over state-of-the-art models.
- 06/21 – 08/21 **Research Intern** at **Theoretical Foundations of AI** under Prof. Debarghya Ghoshdastidar, TU Munich, Germany
Revenue Function of Hierarchical Clustering in Comparison Framework
- Introduced the first framework to evaluate hierarchical clustering directly from comparisons, using a novel revenue function that avoids pairwise similarities.
 - Proposed two average-linkage variants operating on comparisons, achieving a **+0.9 rank correlation** (max: 1) with standard clustering-quality metrics.

🎓 THESIS

- 09/23 – 12/26 **PhD Thesis**, under Prof. Iryna Gurevych, **UKP Lab**, TU Darmstadt
Privacy-Preserving High-Fidelity Synthetic Therapy Session Generation
- Proposed a multi-agent LLM framework for synthetic therapy session generation, achieving **77.2% preference** from human experts across 9 dimensions.
 - Developed a Client Psychological Graph-based framework, achieving a top average rank of **1.6/4** among state-of-the-art baselines and a **4.3% utility improvement** on a counseling benchmark with no safety loss, using Llama3-8B fine-tuned on the generated dataset.
- 08/22 – 04/23 **Master's Thesis**, under Prof. Jiaul Hoque Paik, **CoEAI**, IIT Kharagpur
Universal Transformer for Multimodal Self-supervised Learning
- Proposed a novel crossmodal contrastive loss to train a 12-layer transformer for image-text embeddings on COCO, achieving performance **competitive with Sentence Transformers** on text-only tasks.
- 07/21 – 04/22 **Bachelor's Thesis**, under Prof. Pawan Goyal, **Dept. of CSE**, IIT Kharagpur
Discourse Mutual Information for Dialogue Understanding and Response Generation
- Proposed a novel discourse mutual-information loss for a transformer-based dual encoder.
 - Achieved average improvements of **+4.1%** on dialogue classification and **+10.6%** on dialogue evaluation over state-of-the-art baselines.

🏆 ACHIEVEMENTS

- **ELIZA Scholarship**: ELLIS exchange at INRIA Montpellier (2025)
- **Silver Medal**: Dept. Rank 1, Electronics, IIT Kharagpur (2023)
- **Charpak Lab Scholarship**: Summer internship at INRIA (2022)
- **GKF International Internship Scholarship**: Summer internship at INRIA (2022)
- **DAAD WISE Scholarship**: Summer internship at TU Munich (2021)

⚙️ SKILLS

- **Programming Languages and Softwares**: Python / C++ / C / MATLAB / OpenSmile / OpenFace
- **Libraries**: PyTorch / DeepSpeed / vLLM / HuggingFace / ScikitLearn / Pandas / NumPy

PUBLICATIONS

Graph2Counsel: Clinically Grounded Synthetic Counseling Dialogue Generation from Client Psychological Graphs

Accepted at EMNLP 2026

[Paper](#) | [Code](#) | [Data & Model](#)

Aishik Mandal, Hiba Arnaout, ..., Tanmoy Chakraborty, Iryna Gurevych

MAGneT: Coordinated Multi-Agent Generation of Synthetic Multi-Turn Mental Health Counseling Sessions

Accepted at Findings of EMNLP 2026

[Paper](#) | [Code](#) | [Data & Model](#)

Aishik Mandal, Tanmoy Chakraborty, Iryna Gurevych

SPEAK-SAFE: Secure Processing of Electronic Audio for Knowledge in Suicide Assessment from Therapeutic Exchanges

Frontiers in Digital Health 2026

[Paper](#)

Christopher Landau, Patricia Getty, ..., Aishik Mandal, ..., Andreas Reif, Oliver Grimm

A Comprehensive Survey of Datasets for Clinical Mental Health AI Systems

Preprint

[Paper](#)

Aishik Mandal, Prottay Kumar Adhikary, ..., Iryna Gurevych, Tanmoy Chakraborty

Towards Privacy-aware Mental Health AI Models: Advances, Challenges, and Opportunities

Nature Computational Science

[Paper](#)

Aishik Mandal, Tanmoy Chakraborty, Iryna Gurevych

CaMMT: Benchmarking Culturally Aware Multimodal Machine Translation

Findings of EMNLP 2025

[Paper](#)

Emilio Villa-Cueva, Sholpan Bolatzhanova, ..., Aishik Mandal, ..., Lambebo Tonja, Thamar Solorio

Enhancing Depression Detection via Question-wise Modality Fusion

Oral, CLPsych 2025

[Paper](#) | [Code](#)

Aishik Mandal, Dana Atzil-Slonim, Thamar Solorio, Iryna Gurevych

CVQA: Culturally-diverse Multilingual Visual Question Answering Benchmark

Oral, NeurIPS 2024

[Paper](#)

David Orlando Romero Mogrovejo, Chenyang Lyu, ..., Aishik Mandal, ..., Thamar Solorio, Alham Fikri Aji

A Revenue Function for Comparison-Based Hierarchical Clustering

TMLR, March 2023

[Paper](#) | [Code](#)

Aishik Mandal, Michaël Perrot, Debarghya Ghoshdastidar

Representation Learning for Conversational Data using Discourse Mutual Information Maximization

NAACL 2022

[Paper](#)

Bishal Santra, Sumegh Roychowdhury, Aishik Mandal, ..., Manish Gupta, Pawan Goyal

Knowledge-Aware Neural Networks for Medical Forum Question Classification

CIKM 2021

[Paper](#) | [Code](#)

Soumyadeep Roy, Sudip Chakraborty, Aishik Mandal, ..., Shamik Sural, Niloy Ganguly

TEACHING AND ACADEMIC SERVICE

- 04/25 – 10/25 **Master's Thesis Supervision:** Mateusz Ciesielski, Predicting Personality Domains: A Multimodal Approach.
- 09/23 – 02/24 **Master's Thesis Supervision:** Paulina Kuczykowska, Predicting Depression Severity from Therapy Dialogues.
- 2024 – 2026 **Teaching Assistant,** TU Darmstadt: Ethics for Natural Language Processing.
- 2023 **Teaching Assistant,** IIT Kharagpur: Big Data Processing.
- 2025- **Reviewer:** AAAI AISI Track (2026, 2027), TMLR (2025, 2026), ACL Rolling Review (August 2026).